

YANG YU

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RESEARCH PROFILE

High-resolution regional Earth system modeling; land-atmosphere interaction and hydroclimate; agricultural irrigation, fallowing, dust, and air-quality hazards; coupled atmosphere-ocean modeling; and coastal ocean responses to offshore wind, with extensions to biogeochemical and biosphere-relevant processes.

PROFESSIONAL APPOINTMENTS

2026.05-present	Research Associate III, Woods Hole Oceanographic Institution
2025.05-2026.04	Postdoctoral Investigator, Woods Hole Oceanographic Institution
2022.12-2025.04	Postdoctoral Associate, University of California, Davis
2021.08-2022.11	Postdoctoral Scholar, Institute of Oceanology, Chinese Academy of Sciences

EDUCATION

2016-2021	Ph.D., Environmental Science, Ocean University of China
2014-2016	M.Sc., Environmental Engineering, Ocean University of China
2010-2014	B.Sc., Environmental Science, Ocean University of China

SELECTED RESEARCH ACTIVITIES

- **Land and water management, hydroclimate, and atmospheric hazards (2022-present).** Develop and apply a WRF-irrigation framework to examine how agricultural irrigation modifies soil moisture, humidity, boundary-layer development, near-surface winds, and temperature in California's Central Valley; this framework contributes to the development of a next-generation baseline model for the California Air Resources Board (CARB) air-quality forecasting system.
- **Agricultural fallowing, dust emissions, and air-quality hazards (2024-present).** Develop and apply WRF-Chem and high-resolution land-use data to quantify how fallowing and drought-driven land-surface change affect surface erodibility, dust emissions, visibility, radiation, boundary-layer processes, and particulate pollution.
- **Offshore wind energy and coastal Earth system responses (2025-present).** Develop a coupled WRF-ROMS-Darwin modeling framework to investigate how offshore wind farms modify wind stress, ocean currents, mixing, stratification, nutrient pathways, and biological indicators on the northeastern U.S. continental shelf.
- **Coupled atmosphere-ocean model development and process diagnostics (2021-present).** Develop a ROMS-based hybrid coupled framework that provides methodological foundations in coupled feedback analysis, statistical emulation, and model development for my broader regional Earth system research.
- **High-resolution atmosphere-ocean modeling of extremes and coastal dynamics (2016-2022).** Applied eddy-resolving and cloud-permitting WRF-ROMS simulations to investigate tropical cyclone intensification, remote extreme precipitation, and marginal-sea temperature and salinity variability driven by diurnal surface forcing.

MODELING AND ANALYTICAL SKILLS

Models and modeling frameworks: WRF, WRF-Chem, ROMS, COAWST, and coupled physical-biogeochemical modeling using Darwin and CoSiNE.

Analysis and evaluation: Remote sensing and in situ observations for model evaluation; process-based diagnostics.

Programming and computing: Python, MATLAB, NCL, and HPC workflows.

TEACHING EXPERIENCE

Winter 2025	Teaching Assistant, Numerical Modeling of the Atmosphere, University of California, Davis
Winter 2023	Teaching Assistant, Numerical Modeling of the Atmosphere, University of California, Davis
Assisted students with numerical methods, model setup, coding exercises, and interpretation of atmospheric model output.	

SERVICE AND OUTREACH

2025-present	Seminar Organizer, Department of Physical Oceanography, Woods Hole Oceanographic Institution
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HONORS AND AWARDS

2018	China Scholarship Council (CSC) Scholarship
2018	National Scholarship for Graduate Students, China
2015	Outstanding Graduate Student, Ocean University of China
2015	National Scholarship for Graduate Students, China

PUBLICATIONS

Peer-reviewed journal publications

Bold indicates Yang Yu.

1. Yang, Y., Gao, T., Xu, Y., **Yu, Y.**, Liu, X., Xie, L., Zhang, W., Zhang, B., Guo, X., Long, X., & Zhu, G. (2026). Increasing Socioeconomic Exposure to Extreme Tropical Cyclone Precipitation over China. *Journal of Hydrometeorology*, 27(6), 937-949. <https://doi.org/10.1175/JHM-D-25-0216.1>
2. Lee, A., Chen, S.-H., Huang, C.-C., Nathan, T. R., **Yu, Y.**, Kavaya, M. J., Bedka, K. M., Nehrir, A. R., Thornhill, K. L., Barton-Grimley, R. A., Collins, J. E., Cooney, J. W., Ferrare, R. A., Liu, Z., Lambriksen, B., Schreier, M., Wong, S., & Veals, P. G. (2026). Improved Analysis and Prediction of Tropical Storm Hermine (2022) over the Eastern Tropical Atlantic Using CPEX-CV Observations. *Monthly Weather Review*. <https://doi.org/10.1175/MWR-D-25-0062.1>
3. Li, Y., **Yu, Y.**, Gao, C., Wang, H., & Zhang, R.-H. (2025). Tropical instability waves and their relationships with large-scale oceanic thermal conditions, revealed in ROMS-based hybrid coupled model of the tropical Pacific. *Journal of Oceanology and Limnology*. <https://doi.org/10.1007/s00343-025-5053-9>
4. **Yu, Y.**, Li, Y.-N., Zhang, R.-H., Chen, S.-H., Tseng, Y.-H., Zhang, W., & Wang, H. (2025). A flexible Regional Ocean Modeling System-based hybrid coupled model for El Niño-Southern Oscillation studies - model formulation and performance evaluation. *Geoscientific Model Development*, 18(17), 5527-5547. <https://doi.org/10.5194/gmd-18-5527-2025>
5. **Yu, Y.**, Chen, S.-H., Huang, C.-C., Paw U, K. T., Schmitt, C., Zhao, Z., Suvočarev, K., Kisekka, I., Pyles, R. D., Avise, J., Cai, C., Chang, K.-Y., & Lo, M.-H. (2025). A Numerical Study of the Agricultural Irrigation Effects on Summer Soil Moisture and Near-Surface Meteorology in California's Central Valley. *Journal of Hydrometeorology*, 26(6), 641-659. <https://doi.org/10.1175/JHM-D-24-0129.1>
6. Gong, X., **Yu, Y.**, Shi, X., Lin, X., Liu, G., Dong, Z., Wang, X., Zheng, J., Lembke-Jene, L., & Lohmann, G. (2025). Thresholds in East Asian marginal seas circulation due to deglacial sea level rise. *npj Climate and Atmospheric Science*, 8(1), 83. <https://doi.org/10.1038/s41612-025-00927-y>
7. Zhang, R.-H., Zhang, W., **Yu, Y.**, Li, Y., Tian, F., Gao, C., & Wang, H. (2025). A hybrid coupled model for the tropical Pacific constructed by integrating ROMS with a statistical atmospheric model. *Journal of Oceanology and Limnology*, 43(4), 1037-1055. <https://doi.org/10.1007/s00343-024-4194-6>
8. Zhang, W., Gao, C., Tian, F., **Yu, Y.**, Wang, H., & Zhang, R.-H. (2024). Representing ocean biology-induced heating effects in ROMS-based simulations for the Indo-Pacific Ocean. *Frontiers in Marine Science*, 11, 1-17. <https://doi.org/10.3389/fmars.2024.1473208>
9. Geng, X., Wu, C., Yang, Z., Zhu, J., Tang, K., Lin, J., Liu, Y., Zhang, Y., An, M., Zhao, W., & **Yu, Y.** (2023). Spatiotemporal variations and impact factors of nutrients in the Sanya Bay, northern South China Sea. *Environmental Science and Pollution Research*, 30(31), 76784-76797. <https://doi.org/10.1007/s11356-023-27527-8>
10. Kuo, Y.-C., **Yu, Y.**, & Tseng, Y.-H. (2023). Interannual changes of the summer circulation and hydrology in the East China Sea: A modeling study from 1981 to 2015. *Ocean Modelling*, 181, 102156. <https://doi.org/10.1016/j.ocemod.2022.102156>
11. Zhu, J., Zhou, Q., Zhou, Q., Geng, X., Shi, J., Guo, X., **Yu, Y.**, Yang, Z., & Fan, R. (2023). Interannual variation of coastal upwelling around Hainan Island. *Frontiers in Marine Science*, 10, 1-10. <https://doi.org/10.3389/fmars.2023.1054669>
12. Liu, H., Lin, L., Wang, Y., Du, L., Wang, S., Zhou, P., **Yu, Y.**, Gong, X., & Lu, X. (2022). Reconstruction of Monthly Surface Nutrient Concentrations in the Yellow and Bohai Seas from 2003-2019 Using Machine Learning. *Remote Sensing*, 14(19), 5021. <https://doi.org/10.3390/rs14195021>
13. **Yu, Y.**, Chen, S., Tseng, Y., Foltz, G. R., Zhang, R., & Gao, H. (2022). Impacts of model resolution and ocean coupling on mean and eddy momentum transfer during the rapid intensification of super-typhoon Muifa (2011). *Quarterly Journal of the Royal Meteorological Society*, 148(749), 3639-3659. <https://doi.org/10.1002/qj.4379>
14. **Yu, Y.**, Gao, T., Xie, L., Zhang, R.-H., Zhang, W., Xu, H., et al. (2022). Tropical cyclone over the western Pacific triggers the record-breaking "21/7" extreme rainfall in Henan, central-eastern China. *Environmental Research Letters*, 17(12), 124003. <https://doi.org/10.1088/1748-9326/aca2c4>
15. Meng, S., Gong, X., **Yu, Y.**, Yao, X., Gong, X., Lu, K., et al. (2021). Strengthened ocean-desert process in the North Pacific over the past two decades. *Environmental Research Letters*, 16(2), 024034. <https://doi.org/10.1088/1748-9326/abd96f>
16. Zhang, C., Shi, Z., Zhao, J., Zhang, Y., **Yu, Y.**, Mu, Y., et al. (2021). Impact of air emissions from shipping on marine phytoplankton growth. *Science of The Total Environment*, 769, 145488. <https://doi.org/10.1016/j.scitotenv.2021.145488>
17. Ding, X., Guo, X., Zhang, C., Yao, X., Liu, S., Shi, J., Luo, C., Yu, X., **Yu, Y.**, & Gao, H. (2020). Water conservancy project on the Yellow River modifies the seasonal variation of Chlorophyll-a in the Bohai Sea. *Chemosphere*, 254, 126846. <https://doi.org/10.1016/j.chemosphere.2020.126846>
18. Feng, L., **Yu, Y.**, Gao, H., & Yao, X. (2020). Surface water formation on the natural surface under supersaturation: from local water balance to air pollutant deposition. *Air Quality, Atmosphere & Health*. <https://doi.org/10.1007/s11869-020-00901-y>
19. Qi, J., **Yu, Y.**, Yao, X., Gang, Y., & Gao, H. (2020). Dry deposition fluxes of inorganic nitrogen and phosphorus in atmospheric aerosols over the Marginal Seas and Northwest Pacific. *Atmospheric Research*, 245, 105076. <https://doi.org/10.1016/j.atmosres.2020.105076>
20. **Yu, Y.**, Chen, S.-H., Tseng, Y.-H., Guo, X., Shi, J., Liu, G., Zhang, C., Xu, Y., & Gao, H. (2020). Importance of Diurnal Forcing on the Summer Salinity Variability in the East China Sea. *Journal of Physical Oceanography*, 50(3), 633-653. <https://doi.org/10.1175/JPO-D-19-0200.1>
21. Zhang, C., He, J., Yao, X., Mu, Y., Guo, X., Ding, X., **Yu, Y.**, Shi, J., & Gao, H. (2020). Dynamics of phytoplankton and nutrient uptake following dust additions in the northwest Pacific. *Science of The Total Environment*, 739, 139999. <https://doi.org/10.1016/j.scitotenv.2020.139999>
22. Liu, Q., Jiang, Y., Wang, Q., Wang, M., McMinn, A., Zhao, Q., Xue, C., Wang, X., Dong, J., **Yu, Y.**, Han, Y., & Zhao, J. (2019). Using picoeukaryote communities to indicate the spatial heterogeneity of the Nordic Seas. *Ecological Indicators*, 107, 105582. <https://doi.org/10.1016/j.ecolind.2019.105582>

23. Zhang, C., Yao, X., Chen, Y., Chu, Q., **Yu, Y.**, Shi, J., & Gao, H. (2019). Variations in the phytoplankton community due to dust additions in eutrophication, LNLC and HNLC oceanic zones. *Science of The Total Environment*, 669, 282-293. <https://doi.org/10.1016/j.scitotenv.2019.02.068>
24. Zhu, Y., Li, K., Shen, Y., Gao, Y., Liu, X., **Yu, Y.**, Gao, H., & Yao, X. (2019). New particle formation in the marine atmosphere during seven cruise campaigns. *Atmospheric Chemistry and Physics*, 19(1), 89-113. <https://doi.org/10.5194/acp-19-89-2019>
25. Zhang, C., Gao, H., Yao, X., Shi, Z., Shi, J., **Yu, Y.**, Meng, L., & Guo, X. (2018). Phytoplankton growth response to Asian dust addition in the northwest Pacific Ocean versus the Yellow Sea. *Biogeosciences*, 15(3), 749-765. <https://doi.org/10.5194/bg-15-749-2018>
26. Xu, Z., Song, X., Wang, M., Liu, Q., Jiang, Y., Shao, H., Liu, H., Shi, K., & **Yu, Y.** (2017). Community patterns and temporal variation of picoeukaryotes in response to changes in the Yellow Sea Warm Current. *Journal of Oceanography*, 73(5), 687-699. <https://doi.org/10.1007/s10872-017-0425-1>
27. **Yu, Y.**, Gao, H., Shi, J., Guo, X., & Liu, G. (2017). Diurnal Forcing Induces Variations in Seasonal Temperature and Its Rectification Mechanism in the Eastern Shelf Seas of China. *Journal of Geophysical Research: Oceans*, 122(12), 9870-9888. <https://doi.org/10.1002/2017JC013473>
28. Meng, L., Gao, H. W., **Yu, Y.**, Yao, X. H., Gao, Y., Zhang, C., & Fan, L. (2017). A new approach developed to study variability in North African dust transport routes over the Atlantic during 2001-2015. *Geophysical Research Letters*, 44(19), 10,026-10,035. <https://doi.org/10.1002/2017GL074478>
29. **Yu, Y.**, Gao, H., & Shi, J. (2017). Impacts of Diurnal Forcing on Temperature Simulation in the Shelf Seas of China. *Periodical of Ocean University of China*, 47(4), 106-113. <https://doi.org/10.16441/j.cnki.hdxh.20160138>
30. **Yu, Y.**, Sun, P., Cui, H., Sheng, H., Zhao, F., Tang, Y., & Chen, Z. (2015). Effects of trawl selectivity and genetic parameters on fish body length under long-term trawling. *Journal of Ocean University of China*, 14(5), 835-840. <https://doi.org/10.1007/s11802-015-2885-5>
31. Sun, P., Liang, Z.-L., **Yu, Y.**, Tang, Y.-L., Zhao, F.-F., & Huang, L.-Y. (2015). Trawl selectivity-induced evolution effects on age structure and size-at-age of largehead hairtail (*Trichiurus lepturus*) Linnaeus, 1758 in the East China Sea, China. *Journal of Applied Ichthyology*, 31(4), 657-664. <https://doi.org/10.1111/jai.12774>

Submitted manuscripts

1. **Yu, Y.**, Chen, S.-H., Adebiyi, A., Kibria, M. M., & Pandey, S. (2026). Agricultural Fallowing Drives Extreme Anthropogenic Dust and Visibility Degradation During the October 2021 Dust Event in California's Central Valley. Submitted to *Geophysical Research Letters*. <https://doi.org/10.31223/X50195>
2. Gao, T., **Yu, Y.**, Chen, S.-H., Zhang, W., Gao, J., & Liu, J. (2026). Major contribution of the western Pacific subtropical high to tropical cyclone remote precipitation extremes during the "21.7" Henan event. Submitted to *Journal of Geophysical Research: Atmospheres*.
3. Chen, S.-H., Yang, S.-C., Huang, C.-C., **Yu, Y.**, Matsui, T., Zhu, Y., Fu, B., Xue, X., & Chen, C.-Y. (2026). Influence of aerosol-attenuated backscatter data assimilation on dust analysis and forecasts over the tropical Atlantic Ocean. Submitted to *Quarterly Journal of the Royal Meteorological Society*.
4. Wang, K.-Y., Chen, S.-H., Schmitt, C., **Yu, Y.**, Suvočarev, K., McDonald, I., Chen, X.-M., Di, P., Liu, K., Hus, Y.-K., Thiruvengkatachari, R., & Avise, J. (2026). Impacts of marine air penetration and irrigation on planetary boundary layer height in California's Central Valley. Submitted to *Journal of Geophysical Research: Atmospheres*.

Manuscripts in preparation

1. **Yu, Y.**, et al. A flexible Regional Ocean Modeling System-based hybrid coupled model for El Niño-Southern Oscillation studies. Part II: freshwater flux and salinity feedback.
2. **Yu, Y.**, et al. A flexible Regional Ocean Modeling System-based hybrid coupled model for El Niño-Southern Oscillation studies. Part III: biogeochemical feedback.